Dhruv Aparnathi

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GitHub | Codolio

Professional Summary

Self-driven MERN Stack Developer focused on building responsive and interactive web applications using ReactJS and modern backend technologies. Passionate about frontend architecture, UI/UX design, API development, and solving real-world problems through project-based learning.

Skills

Technical Skills: ReactJS (React 19), Redux Toolkit, JavaScript (ES6+), TypeScript, Tailwind CSS v4, HTML5, CSS3, Node.js, Express.js, REST APIs, MongoDB, Mongoose, Redis, Socket.io, LangChain, RAG, JWT & OAuth 2.0, Razorpay, ImageKit, GSAP, Git, GitHub, Postman, Vite

Non-Technical Skills: Problem Solving, Debugging, Team Collaboration, Quick Learning, Analytical Thinking

Experience

Self-Directed MERN Development — 2024 - Present

- Built frontend and full-stack applications through practical project-based learning and continuous experimentation.
- Developed reusable React components, REST APIs, and responsive interfaces using modern JavaScript technologies.
- Worked with MongoDB and Mongoose for schema design, CRUD operations, and backend integration.
- Improved debugging, state management, and application architecture skills through real-world projects.

How_I_Learned | React Learning Journey — Ongoing

- Created multiple mini-projects to strengthen understanding of hooks, props, component structure, and state management.
- Practiced API integration, frontend optimization, and reusable UI development patterns.

Cohort_Backend — Ongoing

- Building backend-focused applications to improve API structuring, authentication, middleware handling, and database workflows.

Projects

Moodify — AI-Powered Facial Emotion Music Recommender — 2026

- Built a high-fidelity MERN web application matching real-time webcam facial expressions to curated mood playlists.
- Implemented client-side facial emotion scanning (resolving to 6 primary moods) using @mediapipe/tasks-vision and ARKit blendshape standards with a 2.2-second scan interval.
- Designed a custom, fully animated audio player (VibePlayer) with GSAP vinyl turntable rotation, custom sliders, and track listings.
- Secured endpoints with JWT sessions in HttpOnly cookies, email OTP verification via Brevo SMTP API, and Redis caching.
- Developed backend support for bulk audio uploads with automatic audio ID3 metadata parsing and ImageKit cloud hosting.

Tech: React, Vite, Node.js, Express.js, MongoDB, Redis, GSAP, @mediapipe/tasks-vision, Brevo SMTP API, ImageKit, SCSS

GitHub | Live Link

Snitch — Architectural Commerce & Creator Marketplace — 2026

- Architected a full-stack neo-brutalist e-commerce platform and merchant portal featuring kinetic GSAP stagger animations and Lenis inertia scrolling.
- Implemented dynamic multi-attribute variant matrix, real-time inventory validation, and cryptographic Razorpay checkout with HMAC-SHA256 signature verification.
- Secured endpoints with Google OAuth 2.0 dynamic origin resolution, RBAC seller guards, and cross-domain HttpOnly cookie authentication; buffered media uploads to ImageKit CDN.

Tech: React 19, Vite, Tailwind CSS v4, Redux Toolkit, Node.js, Express.js, MongoDB, Razorpay, ImageKit, Passport.js, GSAP, Lenis

GitHub | Live Link

Pax AI — Real-Time AI Search & Q&A Engine — 2026

- Orchestrated real-time, context-aware AI search utilizing LangChain to integrate and switch between Google Gemini and Mistral AI models.
- Implemented Retrieval-Augmented Generation (RAG) using Tavily Search API to fetch, analyze, and synthesize web data into cited answers.
- Established low-latency token streaming and bidirectional communication between the client and Express server using Socket.io.
- Integrated ElevenLabs API for high-fidelity text-to-speech audio playback and ImageKit SDK for secure file attachment hosting.

Tech: React 19, Vite, Tailwind CSS v4, Node.js, Express.js, MongoDB, Socket.io, LangChain, Tavily API, ElevenLabs, ImageKit

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Portfolio

Mac-OS Portfolio Simulation 2025

- Built an interactive, MacOS-inspired developer portfolio featuring draggable window modals, file-system simulation, and a desk-like environment.
- Implemented a custom terminal emulator to allow users to interact with portfolio details programmatically through CLI commands.

Tech: ReactJS, SCSS, react-console-emulator, react-rnd

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Education

B.Tech in Computer Science & Engineering - Parul University Expected 2027

CGPA: 7.4

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Web Development.
- **Learning Experience:** Cohort 2.0 - Sheryians Coding School.

Certifications

- Hackathon Participation Certificate
- Prompt Engineering — Sololearn
- Frontend Development — Sololearn
- Web Development — Sololearn

Achievements

- Successfully built and deployed 10+ React-based frontend and MERN stack projects.
- Consistently expanding full-stack capabilities through continuous coding and practical application builds.